

Multimodal Glioma Segmentation and Classification and NLP Analysis of Scholarly Articles to investigate how to improve Segmentation.

Software Artefacts

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Software Artefacts Link via google drive: https://drive.google.com/drive/folders/1-5LA-4hmf7P0xzP67C415_WjZfcxUKHx?usp=sharing

Jupyter Notebooks, Datasets and Models

- 1) Image Processing Notebook (Image Processing.ipynb):
 - a. Used to convert nii file formats to png (Dataset folder name "Full")
 - b. Used to isolate image with pixel count and sharpness greater than the mean (Dataset folder name "all_images")
 - c. Used to create folders for image classification (Dataset folder name "Training1", "Testing1")
- 2) NLP notebook (NLP.ipynb) :
 - a. Depends on obtaining scholarly articles through pypaperbot ("pip install Pypaperbot" and run desired query to obtain articles containing key words)
 - b. Download wordnet and stopwords
 - c. Used to calculate Unigram, Bigram, Trigram
 - d. Used to identify frequency of expert words
 - e. Dataset folder name Docs/Articles
- 3) Segmentation Notebook (Colab_Brats2019_Segmentation.ipynb) :
 - a. Used to build segmentation models. Dataset required is produced from Image Processing Notebook
 - b. all_images.zip file was used for training
 - c. Segmentation model name "BraTs2019mri_final_seg1.h5"
- 4) Classification Notebook(bratsclass_final) :
 - a. Used to prepare classification models. Dataset required is produced from Image Processing Notebook
 - b. Training1.zip and Testing1.zip was used to train classification model
 - c. Classification model name "brats2019_imagenet_class.h5"
- 5) Patient Analysis(Test on Images.ipynb):
 - a. Used to verify the accuracy of models. Dataset provided was sourced from the university of the west indies
 - b. Patients folder contains all images provided by UWI for testing
- 6) Segment and Analysis Dashboard notebook(Bratsmri_Dashboard.ipynb)
 - a. Requires the installation of Voila in anaconda
 - b. Run the notebook and select voila from within the jupyter notebook to start local server
 - c. Able to analyze patient folders and individual png files
 - d. Review top word frequency year over year
 - e. Review Top expert word frequency year over year

Note: Brats2019 Full dataset can be located at <https://www.kaggle.com/aryashah2k/brain-tumor-segmentation-brats-2019>